

Weekly Report (Pre-registration): Decision Rule for the Zero-Contrast Contour N1

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Notation: the shot-noise sensitivity of CW-ODMR is $\eta = \Delta\nu/(C\sqrt{R})$, where C is the resonance *contrast* (a measured quantity), R the photon count rate and $\Delta\nu$ the linewidth. For power-law responses $C = (1 + \rho)^{-c}$, $R = \Gamma_p(1 + \rho)^{-s}$, $\Delta\nu = (1 + \rho)^w$ with $\rho = \Gamma_p/\Gamma$, the lower-case c, s, w are the *exponents*, and Theorem G states that they enter η only through the single scalar $E \equiv 2c + s + 2w$. What changes sign below is the upper-case C , not the exponent c .

Question

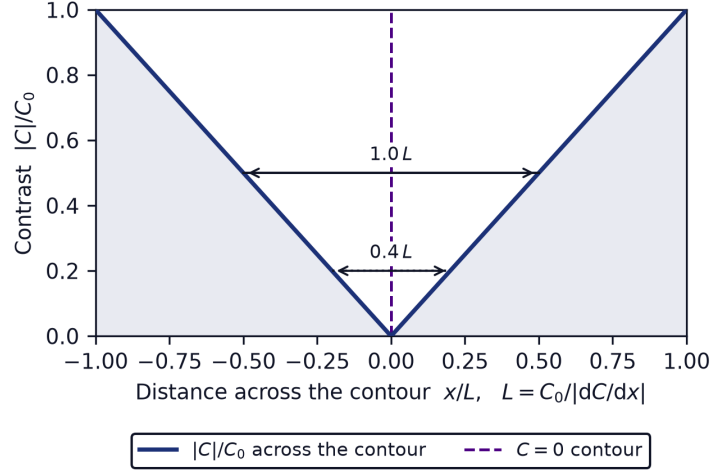
Can a numerical decision rule for identifying the zero-contrast contour N1 as a dark line be fixed before any wide-field ODMR image is opened?

Hypothesis

Along the contour where C crosses zero, $\eta = \Delta\nu/(C\sqrt{R})$ diverges (it is C that reverses sign, not η), so a dark line of vanishing ODMR contrast should appear in the wide-field image.

Evidence / Interpretation

- Bhattacharyya §6.3 reports a complete inversion of the resonance peak above 60 GPa in both [100] and [110], universally across sample geometries [1]. If C passes from positive to negative, a $C = 0$ contour must lie in between.
- Figure 1 fixes what the decision rule has to specify. The apparent width of the dark line is not a property of the contour but of the detection threshold: for a contrast crossing zero with slope $|dC/dx|$, a threshold of $0.5 C_0$ gives a line of width $1.0 L$ and a threshold of $0.2 C_0$ a line of width $0.4 L$, where $L = C_0/|dC/dx|$. A rule that names a width without naming the threshold is empty.



⊠ 1 Contrast magnitude across the $C = 0$ contour, in units of the crossing scale $L = C_0/|dC/dx|$, for a locally linear crossing. The two arrows give the apparent line width at detection thresholds 0.5 and 0.2. Schematic: the crossing slope itself is not predicted by the theory and must be measured.

Gap / Failure

- The threshold, the required continuity of the line and the crossing scale L are not yet fixed numerically, so there is still no criterion separating the feature from instrumental non-uniformity (illumination gradients, the gasket edge, stress concentration). Opening the figure without one would make the judgement post hoc.
- The reported inversion occurs in $[100]/[110]$ and in the gasket region of $[111]$ cut anvils, which is *outside* the domain in which we claim wavelength optimisation ($[111]$ cut culet, $[111]$ NV group, inside the sample chamber). N1 must therefore be stated as a prediction about the exterior of that domain.

Next Step

Fix and register the three numbers the rule needs — detection threshold, minimum line continuity, and the expected range of L — and only then open Fig. 6.7(b).

References

1. P. Bhattacharyya, *Principle and applications of quantum metrology in diamond anvil cells using the Nitrogen Vacancy color center*, PhD thesis, University of California, Berkeley (Fall 2022), §6.3 and Fig. 6.7(b).
2. A. Dréau *et al.*, “Avoiding power broadening in optically detected magnetic resonance of single NV defects for enhanced dc magnetic field sensitivity,” *Phys. Rev. B* **84**, 195204 (2011). Literature grounding for the exponent $E = 3$ of Theorem G.